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Rapid changes in risk attitudes originate from Bayesian inference on parietal magnitude representations

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Risk attitudes—the willingness to accept uncertainty for larger potential rewards—are often treated as a stable individual trait, yet people's risky choices change markedly across contexts and even across repetitions of identical decisions. Here we show, using modelling and neuroimaging, that such rapid changes emerge from Bayesian inference on noisy magnitude representations in parietal cortex. Participants underwent fMRI while choosing between safe and risky options that are either held in working memory or shown on screen. Options held in working memory are noisier and more prone to central-tendency biases, so relatively small payoffs are overestimated and large payoffs underestimated, shifting choices between risk-seeking and risk aversion. A computational model of payoff perception captures these effects mechanistically. Numerical population-receptive-field modelling shows that when intraparietal payoff representations are noisier, choices are less consistent and less risk-neutral. Thus, risk attitudes and their contextual variability can be rooted in perceptual inference on parietal magnitude representations.

People differ widely in their attitudes towards risk. In lab experiments, some people will forgo a lottery with a larger expected value for a substantially lower but certain payout, whereas others will choose the lottery[1–9]. Relatedly, in everyday life, some people are more willing than others to become self-employed, engage in new relationships, or to invest money in the stock market[10]. Neoclassical economic theories describe such individual differences in behaviour using the concept of individual risk preferences: An individual would be *risk-averse* if they prefer to obtain for sure the expected value of a risky prospect (its gain multiplied by its probability) to realising the risky prospect itself, *risk-seeking* if they prefer the risky prospect itself to obtaining its expected value for sure, and *risk-neutral* if they are indifferent between these options. These preferences are believed to be (relatively) stable and generalisable across choice domains[3,4,11]. Preferences are by definition latent and cannot be measured directly; however, they can be inferred from overt behaviour (choices "reveal" preferences, in the terminology of neoclassical economics). Thus, measuring overt risk attitudes in laboratory choices is the standard way to infer underlying risk preferences, which is seen as key for any statements, recommendations, or interventions designed to help people take decisions under uncertainty in a way that increases their chances to meet personal goals.

Empirical data reveal, however, that individual risk attitudes are highly unstable[12,13]. Objectively identical choice sets (i.e., accept/reject specific gambles) can lead to different choices depending on the options that were offered before[14] or in which order choice options are presented[15]. Furthermore, even when the exact same choice problem is presented repeatedly within the same context, decision-makers choose differently across repetitions[7,16,17]. This seems at odds with the idea that these choices mainly reveal stable preferences.

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Adaptations of the preferential framework have tried to account for the stochasticity of risky choice by assuming that preferences are stochastic (Random Utility Theory, RUT)[18] or by assuming that decision-makers may have a preference for responding randomly[19]. Although these theories alleviate the problem of choice inconsistency, they do not specifically address the question of where this randomness comes from and generally do not explain why risk attitudes are so dependent on context (e.g., order of presentation). By contrast, numerous findings in perceptual neuroscience suggest that the degree of randomness in perception is tightly linked to the amount of bias in decisions, suggesting a common origin in perceptual noise[20–26]. This raises the possibility that the noisiness of perception might directly underlie also some aspects of economic attitudes.

In line with this possibility, some models of economic choice, inspired by the perceptual neuroscience literature[16], attribute risk attitudes directly to the noisiness of how decision-makers perceive the choice problems they are faced with[27–31].

These perceptual models of risky choice draw on theories from perceptual neuroscience, framing decisions as fully rational solutions to optimisation problems computed on inherently noisy representations of choice-relevant information. To make optimal choices, our brain makes the best of these noisy signals by applying principles of Bayesian inference[24,32] and/or efficient coding[26,33], which can lead to biases in percepts of the lottery payouts and probabilities. Note that, on average, such biases are beneficial because they reduce the average perceptual error, by employing prior information about the likely causes of sensory signals in case these are noisy and therefore unreliable. For example, there is a well-known positive correlation between the levels of noise in cognitive representations and central tendency (CT) bias[34]: with increased noise, participants tend to regress their percepts of stimulus values towards the mean value of earlier observed stimuli. The Bayesian interpretation of this is that, when noise increases, participants also increasingly revert to their prior beliefs about possible stimulus values—just as normative statistical theories prescribe[16,24].

In the context of risky choices, decision-makers with noisier perception of a choice problem therefore tend to estimate the larger (risky) payoffs downward towards a prior mean, but the smaller (safe) payoffs upwards. As a consequence, decision-makers may underestimate the size of the larger risky payoffs, thereby exhibiting more risk-averse attitudes. This suggests that two people might in fact be similarly risk-neutral, but may show substantially different risk attitudes in their choices simply because they differ in how precisely and thus unbiased they can perceive a risky choice problem.

We have recently provided initial neuroscientific evidence for such a perceptual account of risk attitudes, by showing that there is a link between the noisiness of an individual's neurocognitive magnitude representations in parietal cortex during perceptual judgements and the amount of risk aversion they exhibit in separate economic choices[35]. This link between neural measurements and latent cognitive variables paves the way for asking important questions about the neurocognitive origins of changes in risk attitudes. First, what determines whether individuals show categorically distinct risk attitudes such as risk-seeking, risk-neutral, and risk-averse behaviour[5]? These categories currently still elude perceptual models of risky choice[27,30], which have until now focused on risk aversion. Second, risk attitudes appear highly sensitive to contexts: Choices can differ substantially depending on how decision problems are framed[4], in which order options are presented[15], or which choice options were seen in earlier choice problems[14]. Can we mechanistically account for the influence of context on economic choice, by studying how it alters the perception of the choice-relevant information? Finally, as alluded to above, economic choice behaviour is profoundly variable even without any changes in context[7,16,17]. Where do these apparent fluctuations in risk attitudes come from? Can they originate from endogenous fluctuations of neural noise, in analogy to how such fluctuations affect perceptual decisions about low-level visual features[36]?

To address these questions, we introduce a general computational model of choice, the *Perception and Memory-based Choice Model (PMCM)*. The model builds upon existing perceptual models of risky choice[27,35] that frame risky choice as a Bayesian inference process on noisy percepts of potential payoffs. Critically, in the PMCM, the noisiness of the different percepts relevant to the choice is no longer fixed but depends on the particular context of a specific trial. Specifically, in the experiment presented here, the noisiness of the representation of a choice option may depend on both whether a choice option is perceived directly or held in working memory, as well as on the acuity of neural processing on a particular trial. Thus, the PMCM makes it possible to characterise the psychological and neural mechanisms by which both exogenous environmental contexts and endogenous (neural) factors can shape economic choice on a moment-to-moment basis. This enables us to provide a direct test of perceptual theories of risky-choice mechanisms using biological data.

In the following, we will first describe the risky choice paradigm we developed to modulate the acuity with which choice options are represented in the brain, by manipulating the order in which they are presented. This experimental paradigm also allows us to extract an fMRI signal that corresponds specifically to the neural acuity of the perception of the payoff of the first choice option, independent from any choice-related neural activity that comes later in the trial[35]. To be able to do so, we used the numerical magnitudes of dot clouds to present the payoff magnitudes to the participants, as we have in earlier work[35]. Such stimuli are somewhat different from the Arabic numerals often employed to present potential payoffs; however, seminal work in numerical cognition has established that the neural processing of the numerosity of such stimuli is similar to that of more general (symbolic) numerical information[37,38], including Arabic numerals[35]. Numerosity processing for both types of stimuli engages a more general parietal 'magnitude system'[39], and the acuity of numerosity processing for both types of stimuli is correlated across individuals[35]. However, to establish that our findings are not specific to dot cloud stimuli, we also administered a behavioural version of the task in which payoff magnitudes were presented as Arabic numerals. The key behavioural effects were consistent across both task variants, demonstrating that the observed changes in risk attitudes generalise across numerosity display formats.

Here, we show that rapid changes in risk attitudes arise from Bayesian inference on noisy parietal magnitude representations. We introduce a perceptual model of risky choice—the Perception and Memory-based Choice Model (PMCM)—and demonstrate that it explains why some people are generally risk-seeking while others are risk-averse, and why the same individuals appear risk-averse for some choices but risk-seeking for others. Using fMRI and inverted encoding models[35,40,41], we further show that trial-to-trial fluctuations in the acuity of neural magnitude representations in right intraparietal cortex relate mechanistically to choice consistency and risk attitude in tandem, as predicted by a Bayesian perceptual account. These results indicate that risk attitudes and their variability across trials and contexts originate, at least partly, from perceptual neural processing in parietal cortex. Our Bayesian model captures these links, whereas classical expected-utility models cannot. Our approach and findings have broad implications for economic, psychological, and neuroscience theories of risk-taking, as well as for the paradigms used to study and evaluate risk attitudes.

Results

Experimental paradigm

Thirty participants (14 female, aged 20–34) performed the task twice: Once in a 3 Tesla (T) scanner and once in a 7T scanner. We chose to acquire data on two field strengths to allow for both eye-tracking (3T; data not presented here) as well as ultra-high resolution imaging (7T).

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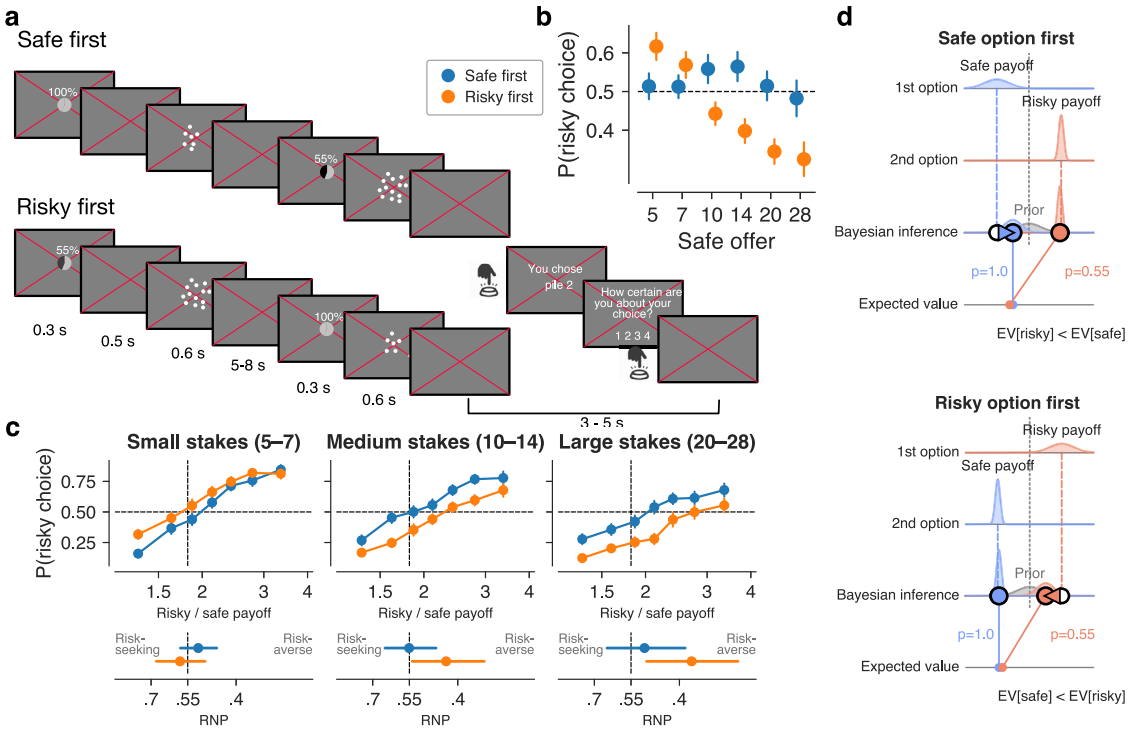


Fig. 1 | Task paradigm and order × stake effects on risky choice. a On each trial, participants chose between a risky option (paid with 55% probability) and a safe option (paid with 100% probability). Prospects were presented sequentially as clouds of 1-CHF coins, each payoff preceded by its probability shown as a pie chart and a percentage. Either option could appear first and therefore had to be held in working memory; every choice set occurred twice per presentation order in every session. **b** Proportion of risky choices as a function of the safe option's magnitude (stake size) and presentation order (hue). Error bars denote ±1 s.e.m. across participants ($n = 30$); the dashed line at 0.5 indicates indifference. **c** Psychophysical curves for three stake sizes (small, medium, large): probability of a risky choice versus the risky/safe payoff ratio, binned per participant (vincentisation) and shown separately per presentation order (hue); shaded error is ±1 s.e.m. across participants. Insets show the risk-neutral probability (RNP; <0.55 risk-averse, >0.55 risk-seeking), estimated with a hierarchical Bayesian probit model and drawn as the 95% credible interval of the group-level posterior; order effects are reported as one-sided posterior probabilities, so no multiple-comparison correction applies. For small-stake gambles participants became more risk-seeking when the risky option was presented first, and more risk-averse for large-stake gambles, consistent with larger central-tendency (CT) effects for remembered than for currently presented payoffs. **d** Schematic of the intuition behind the Perception and Memory-based Choice Model (PMCM): an option presented first is encoded more noisily because it is held in working memory, so the same risky option has a noisier likelihood when presented first (left) than second (right), is pulled further toward the prior and incurs a larger CT bias. Panel **d** is illustrative, using stereotyped parameter values; it shows no data, and no statistical test was performed.

To study how risk attitudes relate to the acuity of perception and the underlying neural representations, we developed an experimental paradigm based on earlier work[35]. In our paradigm, participants had to choose between a sure payoff and a gamble with a 55% probability of winning a larger payoff (Fig. 1a). Crucially, the payoffs of the safe and risky options were presented sequentially; in some trials, the risky options were presented first, in other trials, the safe option was presented first. This design aimed to introduce working memory noise for the first-presented option that had to be memorised, thereby presumably enhancing CT biases for this option.

We kept the payout probability of the risky option fixed at 55%, in line with previous studies[27,35], to streamline the computational modelling of behavioural and neural data. While we recognise the general significance of probability distortion in decision-making[42], our study focussed on the perceptual impacts of payoff evaluations. This focus required a design that emphasised statistical power for these specific effects, and thus fixed payoff probabilities.

The safe option consistently offered one of six possible payoffs, with equal probability but in random order: 5, 7, 10, 14, 20, or 28 CHF. The risky option was determined based on one of eight possible ratios relative to the safe option. These ratios were established through a calibration task conducted outside the scanner, ensuring that the choices presented to participants were non-trivial, as they were near the participants' individual indifference points (see "Methods" for details). Crucially, the payoffs of the safe and risky options were presented sequentially, with in some trials the risky options being presented first, and in other trials the safe option being presented first. This design aimed to introduce working memory noise for the first-presented option that had to be memorised, thereby presumably enhancing CT biases for this option.

In our fMRI experiment, the payoffs were presented as dot clouds (rather than Arabic numerals) because previous work has indicated that such stimulus magnitudes can be reliably decoded from rIPS[35,43]. However, note that we have also performed a very similar experiment using Arabic numerals outside of the scanner (see Supplementary Note 3 for more details).

Different noise contexts shift risk attitudes as predicted by Bayesian inference on noisy payoff representations

We first examined how exogenous factors that modulate neurocognitive noise—in our case, the presentation order of the options—impacted choice behaviour. Psychophysical work on the noisiness of working memory[44–46] suggests that the neurocognitive representation of the initially presented payoffs, retained in working memory, are subject to more noise compared to the options presented subsequently. Thus, taking a perceptual Bayesian perspective[16,24,27,35], we predicted that the options that were presented first should be more sensitive to the central tendency (CT) bias, resulting in overestimation of relatively small payoffs and underestimation of relatively large payoffs. As a consequence, when a relatively small risky payoff was

presented first, participants should be more likely to choose it compared to when it was presented second. Conversely, when a relatively large risky option was presented first, it should be less likely to be chosen. For safe options that were presented first, we expected an analogous pattern - in other words, risk attitudes should be categorically changed by presentation order and overall stake size.

To quantify risk attitudes, we used a psychophysical (probit) model which estimates the indifference point—the ratio between risky and safe payoffs at which individuals show no preference[27,35,47]. Critically, this model disentangles choice consistency from preference, avoiding the confounds inherent in raw choice proportions[47]. The indifference ratio can be converted to a *risk-neutral probability (RNP)*[27], which corresponds to the probability for which a risk-neutral decision-maker would show the same indifference point. Thus, for our paradigm, a RNP of 55% corresponds to risk-neutral behaviour, a RNP lower (or higher) than 55% to a risk-averse (or risk-seeking) attitude. In line with our predictions, our psychophysical modelling showed a strong interaction effect between order and overall stake size on RNP. Specifically, formal model comparisons revealed that the full interaction model received a model weight of 95% using model stacking (a method that finds the optimal linear combination of competing models based on their leave-one-out predictive accuracy[48,49]). Moreover, the expected log pointwise predictive density, ELPD[50] (−5864.7), was at least 22.9 standard errors larger than for the other models; the next best model included only a main effect of stake size (ELPD −6121.4). In line with a CT effect for the first option, when stake sizes were small and the risky option came first, participants were, on average, risk-seeking (average RNP of 58.0% for lowest stakes, 95% CI 48.2–67.3%; see insets Fig. 1c) but became increasingly risk-averse as the average stake size increases (average RNP of 37.3% for highest stakes, CI 25.3–48.6%; there is a significant difference in RNP between the two presentation orders for all stake sizes, all $p_{\text{Bayesian}} < 0.05$).

The Perception and Memory-based Choice Model (PMCM)

To account for more complex risky choice environments, like the one in our experiment, we developed a perceptual model of (risky) choice. The model builds on classic psychophysical models of number perception[51,52] as well as on economic models incorporating these insights to describe risky choice as a perceptual inference process, showing that risk aversion can arise as a result of central tendency effects in perception of the potential payoffs[27,35]. The crucial feature compared to earlier models is that the acuity with which the payoffs of different choice options are represented can dynamically change as a function of exogenous (i.e., choice architecture) or endogenous (i.e., neural/physiological state) context. For example, in our particular experiment, the perception of the first- and second-presented payoffs can be differentially precise, as the noisiness of the neurocognitive payoff representation kept in working memory can be expected to be higher than the representation of the currently perceived payoff[46]. Furthermore, we can also use physiological indices of brain state (e.g., an fMRI activity decoding algorithm) to study how endogenous noise impacts choice behaviour. A second feature of our model is that it allows for different prior expectations about specific choice options based on secondary cues. For example, in most risky choice experiments, the payoffs of risky options are, on average, substantially higher than those of safe options and, given noisy perception, participants could use this information to their benefit. Note that both these features of our model and experiment resemble real-life choice situations, where options are rarely perceived exactly simultaneously and are almost always considered sequentially. By explicitly accounting for this in our design and model, we could test whether and how participants' decisions are affected by this feature of a typical choice situation.

The full PMCM we employed in this study has six parameters: (1) the noisiness with which the first option is perceived (ν_1), (2) the noisiness with which the second option is perceived (ν_2), (3) the mean of the prior on risky payoffs (μ_x), (4) the mean of the prior for the safe payoffs (μ_c), (5) the standard deviation of the prior on risky payoffs (σ_x), and (6) the standard deviation of the prior on safe payoffs (σ_c). Note that these six parameters can be estimated in our experimental paradigm because the noise of the evidence is experimentally manipulated by order, orthogonal to the riskiness of the option. We confirmed the recoverability of the parameters of our model using multiple strategies. When the model was fitted separately to the same participants' data for the two scanner sessions, the six parameters were all correlated across the two sessions, with correlation coefficients $r(28)$ between 0.46 and 0.76 (all $p < 0.05$). A parameter recovery study using the estimated parameters as generating parameters showed substantial correlations between generating and recovered parameters, ranging from 0.74 to 0.93. See Supplementary Note 1 for more details.

Posterior predictive plots confirmed that our model accurately predicts the observed empirical patterns at the group level, including the interaction between magnitudes and order of presentation (Fig. 2a). Moreover, our model also explains highly divergent behaviour across individual participants: Some participants are not influenced by order and/or overall stake sizes, and some participants are risk-seeking rather than risk-averse. Supplementary Fig. 1 gives some examples of typical behavioural patterns of representative participants and their posterior predictive plots.

The first goal of our study was to explain how seemingly categorically different risk attitudes (e.g., risk aversion and risk seeking) may emerge from noisy perceptual inference processes. Our data reveal that there is indeed considerable variability across both participants and choice contexts when it comes to whether behaviour is 'risk-seeking', 'risk-neutral', or 'risk-averse'. Indeed, we found that under some circumstances, our experimental population was risk-seeking on average, namely when the risky option was presented first and the safe option was relatively small (5 or 7; Fig. 1c). Our model-based inference that participants are particularly noisy and/or optimistic about potential prospects can now characterise the latent decision mechanisms and how they mechanistically lead to seemingly different behaviour in specific contexts. This appears more useful than a purely descriptive categorisation based on average risk attitude.

The second main goal of our study was to explain why risk attitudes can shift across different contexts. Our model can naturally explain such context-specific risk attitudes: The risky payoff is particularly overestimated when it is relatively small and noisy. Moreover, it is well-known that a non-negligible fraction of experimental populations shows risk-seeking behaviour on average[5,53]. Current perceptual models of risky choice cannot account for this without assuming additional distortions in probabilities[27,30]. In contrast, our PMCM can account for these effects within one overarching framework: Risk-seeking participants have particularly optimistic prior beliefs about risky payoffs and are particularly risk-seeking when their percepts of the objective risky payoff are noisier (see Fig. 2c; left and middle panel).

One question about our model might be whether it is unnecessarily complex, as it contains relatively many parameters to fit the data. We addressed this question with both qualitative predictive checks and quantitative model comparison techniques, testing whether both dynamic noise and different priors for risky/safe options are strictly necessary to explain the empirical data[50,54]. To do so, we compared the PMCM to four simpler models. The first model was a baseline model where the noisiness of the first and second options were identical and the priors were the same for risky and safe options (model A[27,35]). Note that this model A is equivalent to a standard probit psychophysical model, which models the precision with which decision makers can compare two numerosities (with a potential bias towards one of the two options). In the second model, the noisiness of the first and second options was constant but the priors for risky and

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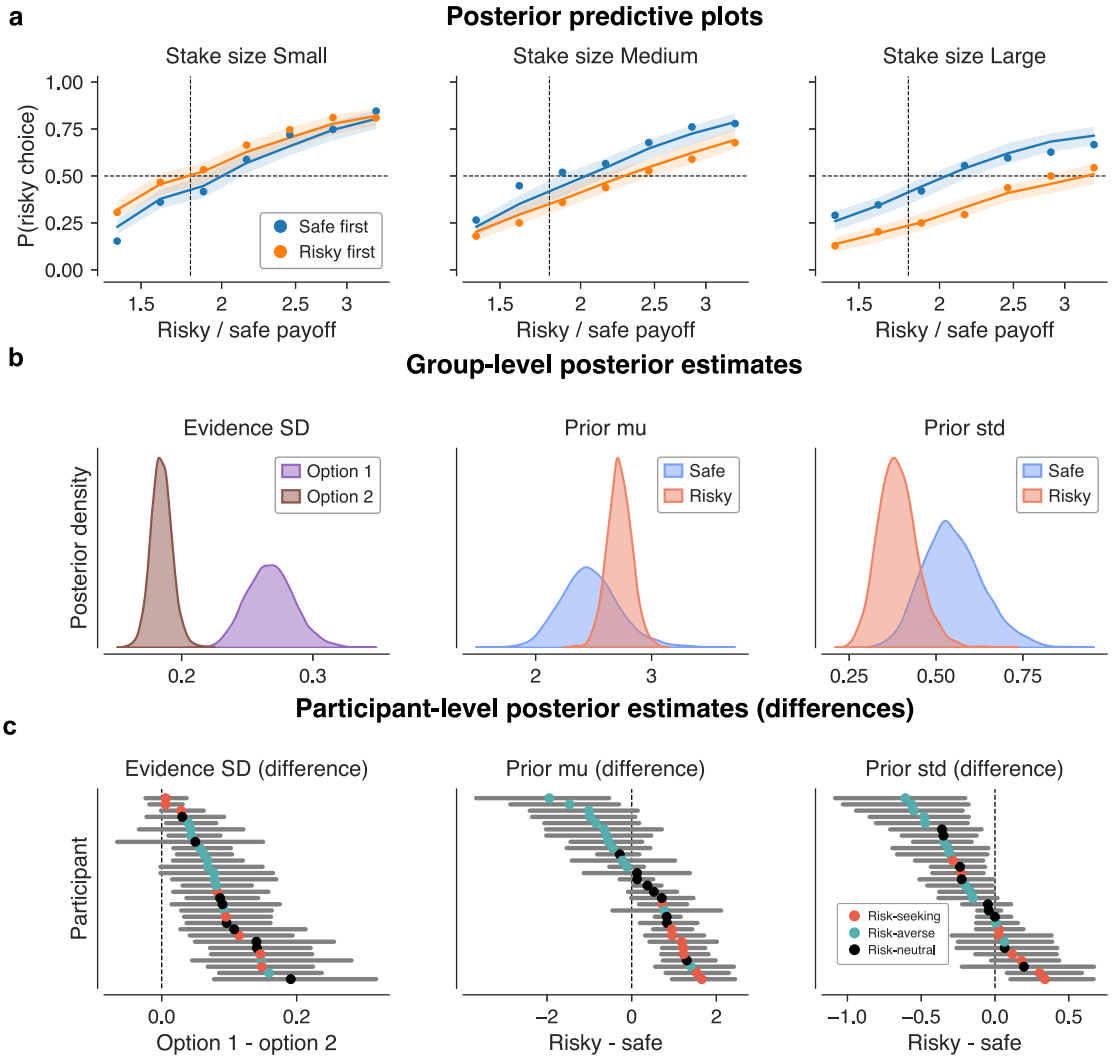


Fig. 2 | The Perception and Memory-based Choice Model (PMCM). Estimates combine the 3T and 7T sessions ($n = 30$). **a** PMCM posterior predictions (line with shaded 95% credible interval) plotted against the group-mean data (markers) for each presentation order and stake size; the predictions closely match the empirical data. **b** Group-level posterior densities (kernel-density estimates) for the model parameters—evidence s.d., prior mean and prior s.d.—coloured by option (Option 1 vs. Option 2 for evidence; Safe vs. Risky for the priors). The first option is encoded with substantially more noise than the second (left), and the prior for risky options has a higher mean (middle) and a narrower range (right) than for safe options, so risky payoffs are more prone to CT effects. **c** Participant-wise posterior estimates of the key parameter differences (Option 1 − Option 2 for evidence s.d.; Risky − Safe for the priors): dots are per-participant posterior means and lines the 95% credible intervals, coloured by each participant's combined risk profile (risk-seeking/risk-averse/risk-neutral). Almost all participants have significantly noisier evidence about the first-presented payoff (left), an effect that also holds at the group level (one-sided posterior probability, <0.001 for both sessions; no correction for multiple comparisons), whereas only a subgroup employs different priors for risky and safe options (middle/right); this difference in priors tracks the participant's risk attitude. See Supplementary Fig. 2 for the two sessions analysed separately.

safe options could be different (model B). In the third model, the noisiness of the first and second options could be different but the prior for the risky and safe options was shared (model C). Finally, we also compared our models to a standard expected utility model (Model D[4,55]). The posterior predictive plots are presented in Fig. 3. Visually, it is evident that none of the alternative models can explain the interaction effect of magnitude and order on the proportion of risky choices. Moreover, a formal model comparison using the Expected Posterior Log Density[50] shows that the PMCM clearly outperforms the other four models. The difference in ELPDs is at least 21 standard errors (Fig. 4). When we used model stacking[48], the model weight of the PMCM was 95%. These measures thus provide evidence for both mechanisms (i.e., varying noise for order and varying priors for riskiness) that are incorporated in our model. To make sure that our model comparison approach was sufficiently powerful, we also simulated 100 data sets using the 5 different models and used the same technique to estimate which was the generating model. When the PMCM was the generating model, it was correctly identified 100/100 times (See Supplementary Note 2 for more details). All in all, we demonstrate that the PMCM makes counterintuitive qualitative predictions that are fully in line with empirical data and that can only be captured by this particular model.

Finally, posterior estimates for the group mean of the model parameters (Fig. 2b) revealed that, as expected, participants had noisier payoff representations for first-presented versus the second-presented option ($p_{\text{Bayesian}} < 0.001$ for both sessions). Notably, while at the group level, the subjective prior for risky payoffs did not have a higher mean than the prior for safe payoffs (3T: $p_{\text{Bayesian}} = 0.446$, 7T: $p_{\text{Bayesian}} = 0.628$, both: $p_{\text{Bayesian}} = 0.156$), there were 13 participants who had a higher mean prior for risky than safe options (95% credible interval of the difference does not overlap with 0, see Fig. 2c). These individual parameter estimates suggest that some participants took

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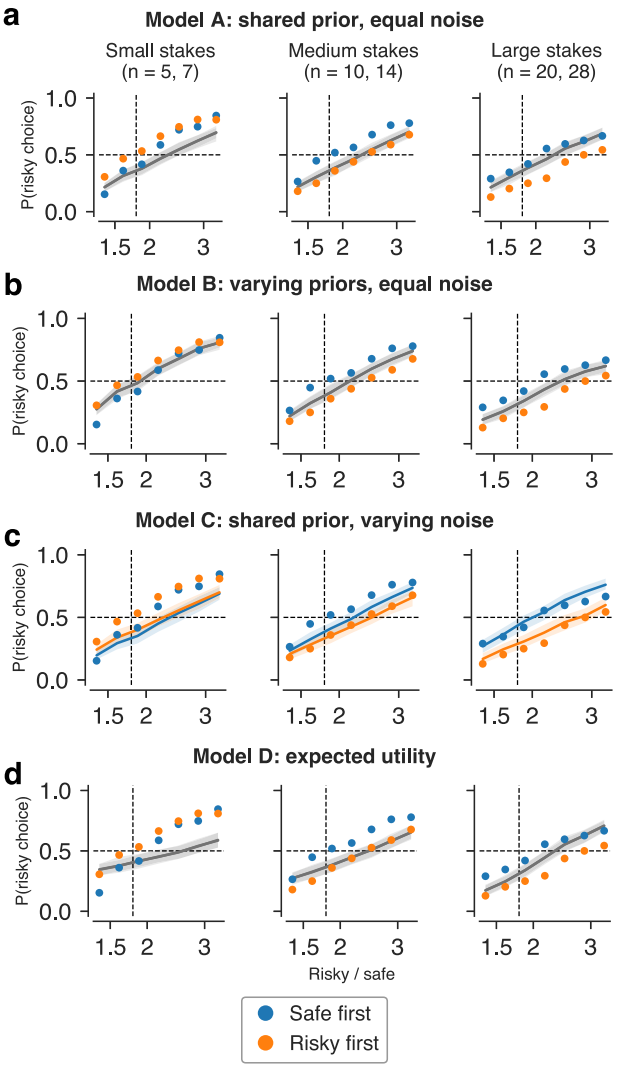


Fig. 3 | Posterior-predictive checks for four alternative models. Each row shows one model and each column a stake size (small, medium, large), for the same cohort as Fig. 2 ($n = 30$). **a** Model A: shared prior, equal noise. **b** Model B: varying priors, equal noise. **c** Model C: shared prior, varying noise. **d** Model D: expected utility. Within each panel, markers are the group-mean data by presentation order, the line is the posterior-predictive mean and the shaded band the 95% credible interval; the dashed vertical line marks the risk-neutral ratio ($1/0.55 \approx 1.82$). Models that cannot distinguish the two presentation orders (Models A, B and D) are plotted in grey; only Model C, which allows order-dependent noise, retains the Safe-first/Risky-first colours. Unlike the PMCM (compare Fig. 2a), none of these models reproduces all qualitative patterns in the data—in particular the order × stake interaction.

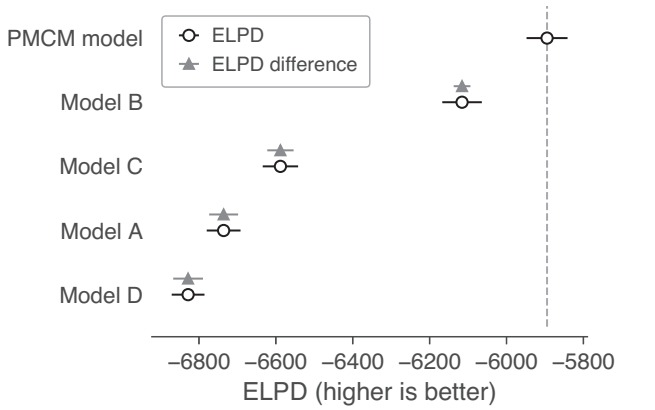


Fig. 4 | Formal model comparison (ELPD). Out-of-sample predictive accuracy for the five behavioural models—the PMCM and the four alternatives (Models A–D) of Fig. 3—quantified by the expected log pointwise predictive density (ELPD) estimated with Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO), and ranked from best (top) to worst ($n = 30$). Open circles show each model's ELPD ± 1 standard error (SE); grey triangles show the ELPD difference ± its standard error (dSE) relative to the best-fitting model, and the dashed vertical line marks the ELPD of the best model. The PMCM has the highest ELPD; its difference to the next-best model is at least 21 times the corresponding dSE.

into account that risky payoffs are higher on average whereas others do not.

In sum, our experimental paradigm and the PMCM can mechanistically characterise how a common and relevant context effect (whether options are held in working memory or perceived immediately) impacts the risk attitude of the participants. The model can explain why the experimental population is risk-averse in some contexts but risk-seeking in others, and it can account for individual differences in risk attitudes.

Decoded payoff representations in parietal cortex predict noisiness and bias in choice

Having established that individual risk attitudes and changes in those attitudes across different choice contexts can be explained by our Bayesian perceptual account of risky choice, we addressed the final question of our study: Where do the trial-to-trial fluctuations in risk attitudes come from? We hypothesised that some choice variability—over and above the variability determined by context and individual differences—can be explained by momentary fluctuations in the acuity of neural representations[56–59]. Our previous work has shown that individual differences in the acuity of parietal magnitude coding predict individual differences in risky choice behaviour: Participants with noisier payoff magnitude representations tend to resort more to their prior beliefs and are, on average, more risk-averse[35]. This paves the way for the question whether there is a comparable relationship between neural noise of choice problems representations and behaviour *within* single decision-makers (in addition to neural noise in putative downstream value computations[60]). Such a relation could in principle be expected based on findings in perceptual neuroscience: For example, trial-to-trial fluctuations in judgements about the orientation of Gabor patches are predicted by fluctuations in orientation representation in primary visual cortex[40,41,61], and judgements about movement directions relate to differences in neural activity in middle temporal area (MT)[62,63].

To be able to test this hypothesis, we first fitted a numerical receptive field (nPRF) model[35,43] to single-trial BOLD responses locked to the *first payoff* presentation, to prevent any decision-related activity from contaminating our measures of neurocognitive representation (Fig. 1a). The nPRF model assumes that patches of cortex are tuned to numerosity—i.e., they have a preferred magnitude to which BOLD responses are largest—and that the size of the response to a stimulus decreases exponentially with its logarithmic distance to the preferred numerosity. Numerically tuned cortical areas were similar to previous studies: The surface-based locations where the model predicted single trial responses best (high explained variance, R^2) largely followed the parietal mask based on our earlier study[35] (see Fig. 5a). Within this mask, vertex-wise R^2 values correlated significantly across the 3T and 7T fMRI sessions within participants with, on average, $r = 0.28$, 95% CI [0.19, 0.38] ($t(29) = 6.2$, $p < 0.001$). The vertex-wise preferred numerosities correlated on average with $r = 0.16$, 95% CI [0.11, 0.21] ($t(29) = 6.0$, $p < 0.001$). These results confirm that the intraparietal sulcus contains participant-specific patterns of numerosity encoding that generalise across sessions and even MRI scanners.

After successfully fitting the nPRF model, we used a Bayesian inversion scheme to decode posterior estimates of presented

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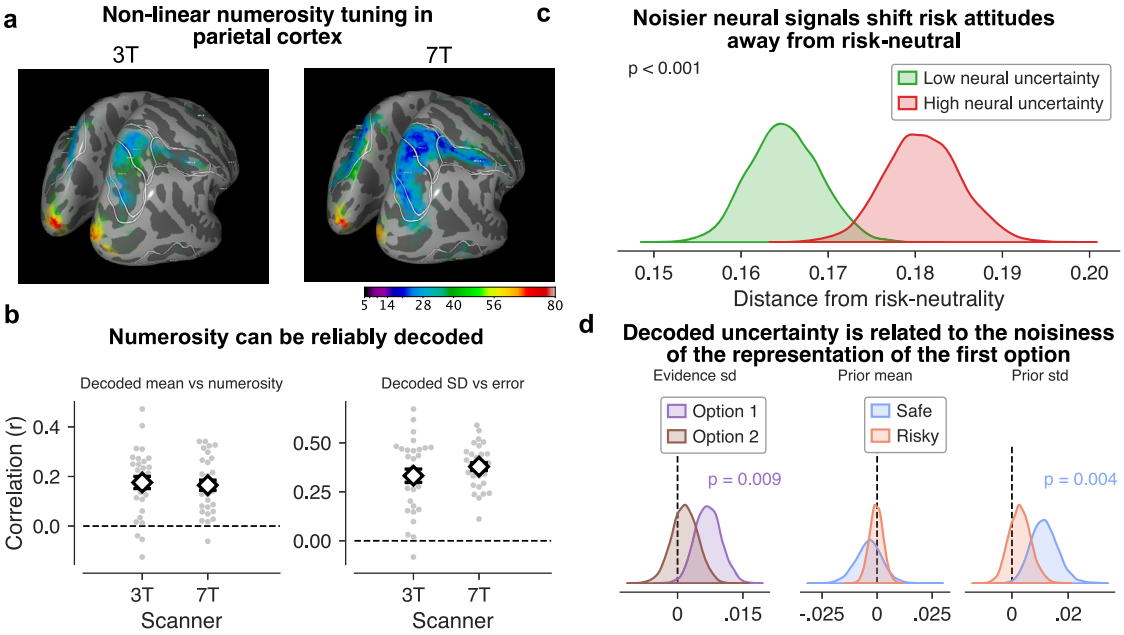


Fig. 5 | Decoded parietal payoff representations predict noisiness and bias in choice. a Numerical population-receptive-field (nPRF) mapping reveals non-linearly tuned, numerosity-sensitive BOLD responses around the intraparietal sulcus, closely following the ROI drawn on an independent data set; shown is the average preferred numerosity across participants in fsaverage space, separately for the 3T and 7T sessions. **b** Per-participant decoding correlations for each scanner (3T, 7T): (i) decoded posterior mean versus objective log-numerosity (decoding accuracy) and (ii) decoded posterior s.d. versus absolute decoding error (uncertainty calibration). Each point is one participant (swarm), the white diamond is the mean ± 1 s.e.m., and the dashed line marks $r = 0$. Nearly all participants show a positive decoding correlation (27/30 at 3T, 29/30 at 7T), and the width of the decoded posterior predicts the objective error of its mean (two-sided one-sample t-tests against zero across participants, $n = 30$; decoding accuracy $t(29) = 7.1$ (3T) and 8.0 (7T); uncertainty calibration $t(29) = 9.8$ (3T) and 18.8 (7T); all $p < 0.001$, uncorrected). **c** Posterior densities of the group-mean distance from risk-neutrality (|RNP − 0.55|; RNP = 0.55 is risk-neutral) for trials with low versus high decoded neural uncertainty (combined sessions). On higher-uncertainty trials choices are more biased—the RNP lies further from risk-neutrality. The annotated value is the one-sided Bayesian posterior probability that the low-uncertainty distance exceeds the high-uncertainty distance ($p_{Bayesian} < 0.001$; per session 3T $p_{Bayesian} = 0.012$, 7T $p_{Bayesian} = 0.013$; no correction for multiple comparisons). **d** Group-level posteriors of the neural (decoded-s.d.) regression coefficient on the PMCM parameters—evidence s.d., prior mean and prior s.d.—coloured by option. The noisiness of the first option (evidence s.d., Option 1) and the dispersion of the prior on safe payoffs increase with decoded neural uncertainty, whereas the second option, the prior means and the prior s.d. on risky payoffs do not. Annotated values are one-sided Bayesian posterior probabilities (no correction for multiple comparisons); evidence s.d., Option 1: $p_{Bayesian} = 0.009$; prior s.d., Safe: $p_{Bayesian} = 0.004$.

numerosities from unseen fMRI data (leave-one-run-out cross-validation)[35]. Note that this method yields the posterior distributions over numerosity given the empirically observed single-trial brain activity pattern. Thus, the analysis does not only give us a point estimate of the numerosity expected to have elicited this BOLD pattern (mean of the posterior) but, crucially, also a measure of the uncertainty surrounding that point estimate. Earlier work in perceptual neuroscience has shown that this posterior uncertainty surrounding a stimulus feature decoded from brain activity predicts the randomness and bias of choices and judgements on those very stimulus features[35,40,41].

Before linking neural uncertainty to behaviour, we first assessed the acuity of the decoder. The presented payoff magnitudes were decoded well above chance on the level of single trials, with an average correlation between actual numerosity and decoded numerosity of $r = 0.176$ (3T; $t(29) = 7.1$; $p < 0.001$; 95% CI [0.13, 0.23]) and $r = 0.165$ (7T; $t(29) = 8.0$, $p < 0.001$; 95% CI [0.12, 0.21], see also Fig. 5b). Importantly, the decoded neural uncertainty (s.d. of the decoded posterior) was significantly correlated with the error of the decoder (absolute difference between mean posterior and actual numerosity) with $r = 0.332$ (3T; $t(29) = 9.8$, $p < 0.001$; 95% CI [0.26, 0.40]) and $r = 0.379$ (7T; $t(29) = 18.8$, $p < 0.001$, 95% CI [0.34, 0.42]). Since the uncertainty of the decoder relates directly to how well it can actually decode a given neural pattern, we can be confident that the variance in the decoded posteriors is meaningful and a useful proxy for the neural noisiness of the payoff magnitude representations that the decision-makers base their choices on[35,40,41].

We also conceptually replicated an important result reported before by ourselves[35] and others[64], namely that participants with particularly noisy neural representations of magnitudes in the right parietal cortex also show noisier and biased behaviour. Specifically, participants who had a higher correlation between neurally decoded numerosities and the presented numerosities tended to have a less noisy representation of the first stimulus magnitude according to our behavioural PMCM (the correlation between decoding correlation and the dispersion of the likelihood of the first option, υ_1, $r(28) = -0.500$, $p = 0.005$, 95% CI [−0.73, −0.17] for 3T and $r(28) = -0.406$, $p = 0.026$ for 7T, 95% CI [−0.67, −0.05]). Importantly, participants for which we decoded posteriors with, on average, a higher standard deviation also had a larger υ_1-parameter according to the PMCM ($r(28) = 0.461$, $p = 0.010$, 95% CI [0.12, 0.70] for 3T and $r(28) = 0.406$, $p = 0.026$, 95% CI [0.05, 0.67] in the 7T data), thus replicating our earlier findings[35] twice, using different scanners and field strengths.

After reconfirming with these replications both the robustness of our IPS-signal decoder and the relevance of these parietal regions for individual risk attitudes, we proceeded to test whether *within each participant*, trial-to-trial fluctuations in apparent risk attitudes are related to latent fluctuations in the neural code that represents the potential payoffs that are at stake. To test this hypothesis carefully, we broke it down to two subhypotheses: On trials where neural payoff representations are particularly noisy, choice behaviour should (1) become less consistent and (2) more biased towards prior beliefs. In the PMCM, both effects can jointly be explained by an increase in noise. However, we aimed to first test both hypotheses separately. To

do so, we used a standard psychophysical model that fits a psychophysical curve with both an indifference point and a psychometric slope to the choice data as a function of the log-ratio of the risky and safe payoffs[27,35]. To make sure we found an effect of neural uncertainty over and above any order- and magnitude-effects, we performed a median split separately for every possible first-presented payoff and fitted separate psychometric curves to these two sets of trials that were matched in payoffs and order but showed relatively high or low neural noise.

In line with our expectations, the psychophysical slope was substantially shallower for trials with a higher decoded neural uncertainty ($p_{Bayesian} = 0.046$ for the 3T data set; $p_{Bayesian} = 0.024$ for the 7T data set). Thus, the noisier the signal was according to our neural decoder, the more randomly (i.e., independent from the presented payoffs) participants chose between the options. To test whether participants were more biased in their perception of the potential payoffs when their neural representations were particularly noisy, we tested whether estimated indifference points were further away from risk-neutrality when neural noise was higher. To quantify risk-neutrality, we again used the *risk-neutral probability* (RNP). We fitted a hierarchical Bayesian psychometric model and sampled the resulting participant-, payoff- and order-specific RNPs. These RNPs were indeed farther away from risk-neutrality (55%) when the neural uncertainty was higher (18.5 percentage points, s.d. 9.2 versus 16.7 percentage points, s.d. 9.2; $p_{Bayesian} < 0.001$, Fig. 5c; these results replicate when separately estimated for two sessions, $p_{Bayesian} = 0.012$ for 3T and $p_{Bayesian} = 0.013$ for 7T). Thus, the less precise the numerosity-tuned responses in IPS according to our decoder, the more participants reverted to their prior beliefs and the less risk-neutral they became in their risk attitudes.

Finally, as the ultimate test of the hypothesis that trial-to-trial fluctuations in risk attitudes are (partly) driven by fluctuations in neural noise, we tested whether an increase in the noise parameter of the PMCM can explain the observed link between neural acuity and behavioural consistency and choice. We hypothesised that, in terms of PMCM parameters, the first choice option should be represented more noisily when the corresponding decoded signal was less precise. Therefore, we refitted the PMCM to behavioural data, but now all six model parameters were estimated as a linear sum of a (participant-wise) intercept and a regression parameter that was multiplied with the *z*-scored trialwise neural uncertainty measures (i.e., the standard deviation of the decoded posterior). If this latter regression parameter differed from 0, the corresponding latent variable linearly varied with neural uncertainty[65,66]. This analysis confirmed that both the noisiness of the first option (3T: $p_{Bayesian} = 0.043$; 7T: $p_{Bayesian} = 0.044$; both: $p_{Bayesian} = 0.009$; see also Fig. 5d and Supplementary Fig. 3) and the standard deviation of the prior on safe payoffs (3T: $p_{Bayesian} = 0.042$, 7T, $p_{Bayesian} = 0.016$; both: $p_{Bayesian} = 0.004$; see Fig. 5d and Supplementary Fig. 3) linearly increased with increased neural uncertainty. These results align with the psychophysical modelling results: When neural noise is high, the representation of the payoff of the first option becomes noisier, resulting in larger CT biases and more variable behaviour. However, the increased dispersion of the prior belief about safe payoff magnitudes with increased uncertainty reveals that it is mostly the risky (rather than safe) options that drive the increased CT bias with increased neural uncertainty.

In conclusion, we successfully decoded payoff magnitude representations during economic choice from right parietal cortex and replicated earlier findings of a link between the acuity of an individual's numerical magnitude representation in the IPS and their choice consistency and risk attitudes. Crucially, we now found that even trial-to-trial fluctuations in the acuity of payoff representations in parietal cortex were mechanistically linked to trial-to-trial fluctuations in choice consistency and risk attitude. This provides direct biological evidence that the endogenously fluctuating precision of neural magnitude representations underlies time-variations in revealed risk attitude.

Choices in symbolic presentation format also reveal shifts in risk attitude as a result of order and stake size

In the fMRI experiment, we used dot clouds to present potential payoffs, because such payoff stimuli gave us the most reliable signal to probe neurocognitive representations in the parietal lobe using fMRI[35,43]. One might wonder, however, if the observed effects of presentation order and their interaction with stake size might be specific to this non-symbolic presentation format of dot clouds. The non-symbolic presentation format we used may also introduce some stimulus ambiguity that might be weaker in more traditional economic choice tasks using Arabic numerals (although there can be profound noise and bias in the perception of symbolic numerical magnitudes as well[37,67,68]). To address these concerns, we conducted an additional experiment with 58 participants (average age 24.6, range 20–41; 31 females) who performed a very similar choice task, but now the dot clouds were replaced with Arabic numerals (e.g., "CHF 20.10" or "CHF 10.08"; see Fig. 6a).

Just as for the fMRI data, we fitted a set of psychophysical choice models, where the probability of choosing the risky option is a (probit) function of the ratio of the risky and safe option. Choice consistency and risk attitude were modelled separately for different stake sizes. Formal model comparison again identified as the best model a full interaction model in which the indifference point (i.e., the risk-neutral probability, RNP) corresponded to even more risk aversion when risky options are presented first *and* they are relatively large (ELPD: −5383.73 for *order x stake* model vs. −5430.28 for *stake* model, dSE: 11.09; see Fig. 6b-d). Thus, this additional analysis is consistent with the behavioural results from the fMRI study: Participants become more risk-averse when risky options are presented first (and are therefore represented more noisily), in particular when stake sizes are large (and the noisy magnitude representations are biased more towards the smaller prior). This is also consistent with the interpretation of the CT account that risky options are underestimated relatively more when they are objectively large. Additionally, our results also replicate earlier findings[27,35] that individual differences in choice consistency and indifference point (risk attitude) are significantly correlated ($r(56) = 0.344$, $p = 0.008$, 95% CI [0.09, 0.55]). This is again in line with the central assumption of the perceptual account of risk attitudes: Participants whose choices are based on noisier representations (and therefore more inconsistent) also deviate more strongly from risk-neutral behaviour.

To account for the results more mechanistically, we also fitted the four different versions of the PMCM, with varying or shared priors for risky and safe payoffs, as well as different or equal amounts of noise for the 1st and 2nd presented option. Formal model comparison revealed that, just as in the fMRI study, only the full PMCM model could explain the empirical data and clearly outperformed a standard expected utility model; see Supplementary Note 3 for more details. Ten out of the 58 participants robustly showed more noise for the 1st compared to the 2nd option, in line with working memory noise on the first-presented option. However, surprisingly, a smaller number of participants (5/58) robustly ($p_{Bayesian} < 0.05$) showed more noise for the 2nd option than the 1st option, contrary to a memory noise effect. We believe these results might reflect differences in attentional strategies[69], which could be investigated in future modelling work using eye tracking and/or choice-specific noise manipulations.

Taking stock, the key finding of this additional experiment is that the order in which options are presented shapes risk attitudes, and this effect interacts with overall stake size—even when using more traditional, symbolic stimuli to convey payoff magnitude. These results show that presentation order effects—and their interaction with stake size—are not specific to dot cloud displays but also generalise to more natural presentation formats. Moreover, the observed data using Arabic numerals are well explained by our PMCM, whereas earlier perceptual models of risky choice and

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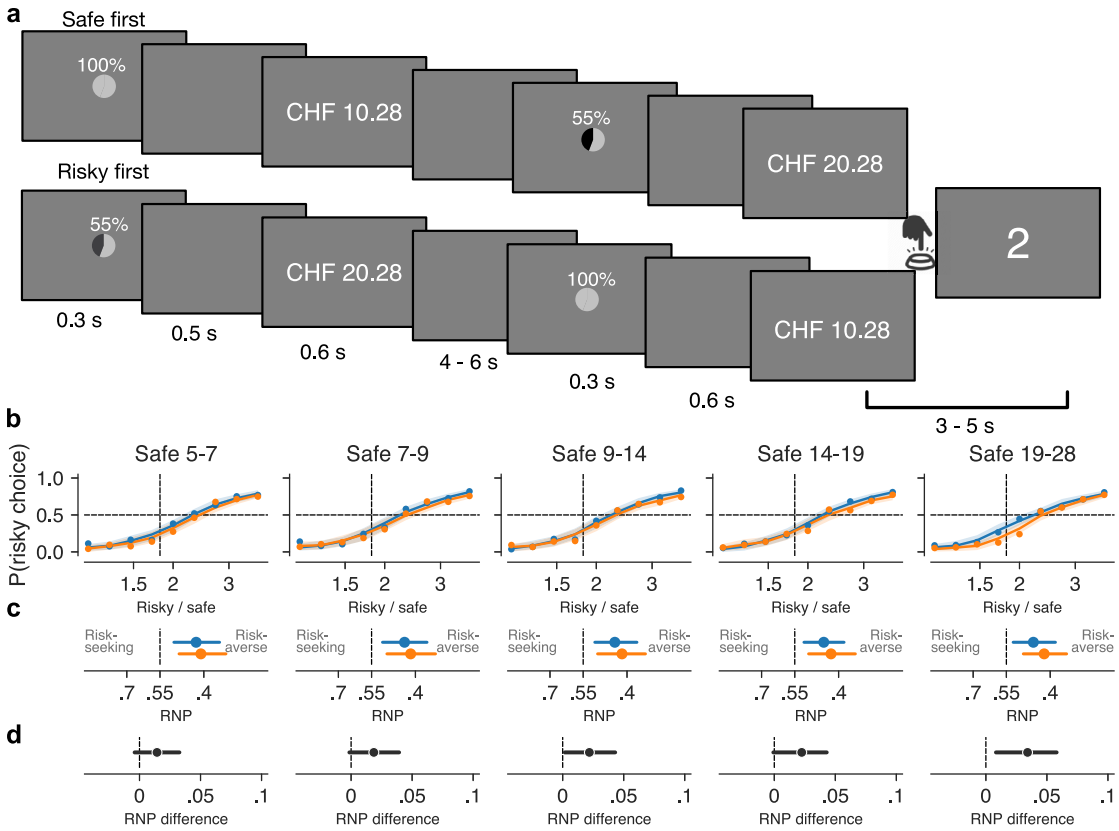


Fig. 6 | Symbolic (Arabic-numeral) experiment. a Additional behavioural paradigm in which potential payoffs were presented as Arabic numerals (e.g., 'CHF 20.28') rather than dot clouds ($n = 58$). **b** Psychophysical curves for five stake bins (safe payoff 5–7, 7–9, 9–14, 14–19 and 19–28): the probability of a risky choice as a function of the risky/safe payoff ratio, separately for each presentation order (hue). Dots are the raw data and the line with shaded band the posterior-predictive mean with 95% credible interval from a hierarchical probit model; the dashed horizontal line marks indifference (0.5) and the dashed vertical line the risk-neutral ratio (1/0.55 ≈ 1.82). **c** Risk-neutral probability (RNP) for each presentation order and stake bin (point = posterior mean, line = 95% credible interval); RNP = 0.55 (dashed line) is risk-neutral, with lower values indicating more risk-averse and higher values more risk-seeking behaviour. **d** Difference in RNP between the two presentation orders for each stake bin (point = posterior mean, line = 95% credible interval; dashed line at 0). Participants were more risk-averse when the risky option was presented first, an effect that grew with stake size.

standard expected utility frameworks fail to capture the full pattern of results. This provides strong evidence—consistent with earlier work[27,35]—that perceptual processes and Bayesian inference play a key role in risky choice, even when information is presented in seemingly unambiguous, symbolic form. See Supplementary Note 3 for more details on this additional experiment.

Discussion

Overt risk attitudes—and the latent underlying risk preferences—play a central role in many aspects of our behaviour, but we still don't completely understand where they come from and why they change across different contexts or even repetitions of identical choices. Here, we provide modelling and neurobiological evidence that rapid changes in individual risk attitudes can originate in the neurocognitive processes that underlie our perception of (monetary payoff) magnitudes. We introduced an experimental paradigm that modulates the noisiness with which different choice options are perceived, by presenting them in different orders, as is typical for everyday choice situations in which options are considered one after another. This sequential presentation made choice options more/less prone to working memory noise[45,46], resulting in central tendency effects[24,34] that led participants to perceive the very same choice options as larger or smaller depending on their presentation order, magnitude, and riskiness (reflected in different priors for risky and safe options). We captured these effects with a Bayesian model of risky choice—the PMCM—that explicitly accounts for dynamic fluctuations in the relative noisiness with which different choice options are perceived, as well as for different prior expectations based on the context of a particular choice problem. Moreover, by using fMRI and inverted encoding models, we could provide neural evidence for such rapid fluctuations in neurocognitive noise, showing that these predict risky choice in a manner consistent with the PMCM.

Our results have important implications for both theoretical and empirical work on risk attitudes. First, in line with other recent work[13,14,17], they defy the conventional economic wisdom that risk attitudes are stable[2] and demonstrate that they can systematically vary due to fluctuations in the noisiness of neurocognitive representations of the relevant decision variables. This relevance of noise in the decision-making process for risk attitudes dovetails with earlier work highlighting that differences in risk attitude may easily be conflated with differences in choice consistency[47,70,71]. However, rather than assuming that noise may affect choices unsystematically, at a late stage of the decision process, our model emphasises that noise originates early in the decision process, during perception of the choice-relevant information, thereby determining risk attitudes already before any actions are taken. These perceptual influences on overt behaviour pose challenges to the common approach of interpreting choices as revealing (only) latent risk preferences.

A second key implication of our study—congruent with other recent work[27,72,73]—is that a large part of inter- and intra-individual variability in risky choice behaviour can be explained by fluctuations in basic perceptual neurocognitive representations, over and above

any possible differences in subjective valuation of the choice options. We show that decision-makers can seemingly reverse their risk attitude (i.e., become risk-seeking rather than risk-averse) simply due to changes in the order in which choice options are presented. This raises the crucial question to what extent variability in risky choices really reflects subjective valuation (rather than noisy perception), as usually assumed in standard economic models. Notably, our PMCM, which contains no subjective valuation process, could explain a wide range of choice patterns across subjects and contexts, whereas a standard economic choice model based exclusively on subjective evaluation (the EU model) failed to do so. While this evidently does not prove that subjective valuation is not involved in risky choices, it still suggests that perception may play an important role and needs to be accounted for explicitly when researchers attempt to measure risk preferences, at least in the small-stake laboratory experiments widely used in the field.

This substantial impact of perception on risk attitudes (rather than valuation) might explain the 'risk elicitation puzzle': the fact that risk attitudes (or "elicited preferences") are only marginally correlated across elicitation methods[6]. Moreover, the central role of perceptual effects in risk attitudes may have important implications for welfare economics, which often relies on elicited preferences to calibrate models of the impact of economic policies[74]. Thus, future work should attempt to further unravel potentially separate contributions of neurocognitive processes related to subjective valuation versus perception to individual preferences and, if possible, develop economic choice tasks where perception can be dissociated from participant's subjective valuation.

Future studies attempting to dissociate valuation and perception should systematically manipulate both sets of processes. Valuation may be experimentally manipulated by changing objective choice properties and comparing qualitatively different decision-makers, while perception may be influenced by orthogonally manipulating the acuity of the choice-relevant information. Here we did so by representing potential payoffs with non-symbolic stimulus arrays and changing their order.

One might argue that this choice context may induce effects in a purely 'low-level perceptual' manner that should be absent for choices with more common Arabic numerals. However, there is ample evidence that the perception of Arabic numerals is prone to similar perceptual effects[68]. For example, when participants have to make decisions about rapidly-presented Arabic numerals, they are sensitive to order effects[67] and the underlying statistics of the presented numerals[75], just as for the non-symbolic presentation formats in the study presented here. Furthermore, recent neuroscience studies have shown that numerosity-tuned regions similarly represent both symbolic Arabic numerals and non-symbolic stimulus arrays[76,77]. In line with this, our recent work[35] has shown that the performance on a perceptual task involving the numerosity of stimulus arrays similarly predicts choice consistency and risk attitude in risky choice tasks that present payoffs either non-symbolically or as Arabic numerals. Finally, in the present study, we also presented a choice task where the non-symbolic stimulus arrays were replaced with unambiguous, Arabic numerals. In that experiment, we also found an effect of order on risk attitude, as well as an interaction with stake size. A closer look at the behaviour and parameter estimates of individual participants showed a wide array of potential choice behaviour that could not be explained by a standard expected utility model (which was also confirmed with formal model comparison measures). All this highlights that choice problems presented in non-symbolic and symbolic presentation formats engage similar neural magnitude representations, and that our model can capture common processes underlying choices about payoffs presented in both presentation formats.

The findings that our Bayesian model of the inference underlying perception could account for several counterintuitive context effects on risk attitudes, for choice sets both ambiguous and non-ambiguous payoff stimuli, suggests that this type of model may be important for understanding several other effects documented in the literature. For example, another well-established determinant of the acuity of choice information is the time that choice information is attended to. Indeed, eye-tracking data show that both the order and duration with which participants attend to relevant choice variables profoundly influence economic choice[78]. Importantly, participants act *as if* they multiply relative values of choice options with the amount of time they gaze upon a choice option[79,80]. This multiplicative effect is fully consistent with a Bayesian account where attention increases the acuity with which we perceive information. Thus, particularly valuable options are chosen more often with increased attention, but less-valuable options less often (since the CT effect will decrease as a function of dwell time)[81]. Another way in which perceptual processes preceding putative valuation may affect risk attitudes are changes in the statistics of the local environment[82]. Indeed, it is well-known that economic behaviour is sensitive to the larger choice set across an experiment[14]. Future studies should thus attempt to further test with rigorous Bayesian modelling whether these choice set effects are not only qualitatively but also quantitatively consistent with a model where subjective priors adapt to the local choice context[24,30,83].

Our work also has implications for existing neuroeconomic studies and opens up several research directions. First, studies attempting to identify neural correlates of subjective value[84] need to take into account that at least under some conditions, subjective values inferred from choice behaviour might be a compound measure that also includes perceptual noise and biases, which can profoundly influence any decision variable constructed from this information. This means that neural correlates of subjective value might include perceptual areas as well as 'true' valuation areas[85,86]. However, our work also suggests that computational modelling of behavioural and neural data could help to decompose these sources of choice variability[87]. It will be interesting to see whether explicitly incorporating perception in such models can help to elucidate whether subjective value is represented in the brain completely independently from perceptual variables.

Second, our results suggest a key role for parietal regions in economic decisions that involve (numerical) magnitudes. Most neuroeconomic theories propose that, during choice, subjective values are represented in the ventromedial and/or orbitofrontal cortex (vmPFC/OFC)[88], whereas parietal cortex is generally conceptualised as a brain region that integrates the final evidence for different choice options based on subjective or objective stimulus properties[89–91]. However, our results suggest that, at least when magnitudes are involved, parietal cortex also directly represents initially choice-relevant stimulus information during perception of the choice problem. This provides an explanation for earlier discoveries of neural substrates of subjective value of specific choice options in the parietal cortex of non-human primates[92,93], as well as recent work using brain stimulation that shows that activity in parietal cortex is causally involved in risky choice[94,95].

Clearly, the precise role of parietal cortex in economic choice remains to be fully elucidated and is likely to depend on the specifics of the choice task, as well as the phase of decision-making during which one monitors parietal activity. However, one promising working hypothesis could be that when objective magnitudes are involved, parietal cortex can both represent relevant magnitudes (e.g., probability of payoff[96] or monetary amounts[35]) of the option currently under consideration/attention[96,97], as well as accumulate evidence for choice options as also found in recent work on perception[98]. Notably, almost all work in (human) neuroeconomics has assumed linear coding of subjective value in large cortical locations that adhere to macro-anatomy (i.e., MNI space). However, our work suggests a non-linear code that is only loosely related to macro-anatomy—at least for parietal representations of potential payoffs (see also work on non-linear

coding of probabilities[96]). More sophisticated fMRI analysis approaches may thus be needed to fully delineate representations of objective stimulus information and subjective values during economic choices[86]. For instance, studies decomposing subjective value in its constituent parts (basic perceptual and subjective) should aim to use analysis frameworks that allow for non-linear codes and take into account individual functional neuroanatomy. An exciting open question for such work may be how fluctuations in early perceptual representations of a choice problem (e.g., potential payoff) covary with prefrontal representations of subjective value.

A central mechanism in Bayesian theories of (economic) cognition are prior beliefs about the state of the world. Some have argued that this may make Bayesian models of cognition overly flexible[99] (but see ref. 100). In the PMCM, decision-makers have prior beliefs over the plausible ranges of risky and safe payoffs. Because the mean and spread of these priors are free parameters, the PMCM is indeed a relatively flexible model. However, both formal model comparison and careful qualitative inspection of the empirical data show that less flexible models of risky choice (both of perceptual and preferential nature) can simply not account for the qualitative behavioural patterns in our data. Having said that, future work should validate whether change in the objective distributions of payoffs will result in the predicted behavioural changes and corresponding parameter shifts of the PMCM. Encouragingly, earlier work on both risk[30] and loss aversion[14] has already shown that human decision-makers are sensitive to the statistical properties of earlier choice problems, in line with Bayesian theories of economic choice. An important, related question is how, and on which time scale, human decision-makers can learn and adapt their subjective priors. After some initial attempts at including such a learning mechanism in our model, we believe that the choice paradigm we employed here is not suitable for such questions, because the amount of information present in a single accept/reject decision is limited. Therefore, future work might employ different paradigms with a more continuous behavioural measure, such as willingness-to-pay[101].

A related critique of Bayesian theories, and potentially the PMCM, is that they are not mechanistic[99] models of neural function but rather prescriptive models of information processing. Nevertheless, the PMCM does give us an increased understanding of the mechanism by which risk attitudes arise, through the lens of Bayesian perceptual inference and its link to neural activity. Concretely, without an understanding of working memory noise and the formalisation of its effects using the Bayesian framework, it remains very challenging to explain the rather complex observed stake x order interaction. Moreover, our neurocomputational modelling results link the abstract Bayesian formalism to concrete, neural signals that we observe using neuroimaging: Payoff magnitudes are represented in a non-linearly tuned population code in parietal cortex, which correlates with a posterior belief. The less consistent this population code across its subpopulations, the more uncertain the decision-maker's beliefs should be[32] and the more they should revert to their prior beliefs. Exactly this link between neural and behavioural signals, which forms a concrete decision-making mechanism, is what we observed in our data. While much work remains, our findings show how abstract, formal Bayesian theories of cognition can be integrated with measurements of corresponding neural population codes, which offers an interesting approach for gaining a more mechanistic understanding of decision-making.

Our work aligns with a recent body of research that conceptualises economic choice idiosyncrasies as rational responses to cognitive capacity constraints[16,25,72,102,103] and makes three contributions to this literature. First, we present a computational model that can be used to estimate how choice contexts modulate both prior beliefs and perceptual noise associated with different choice options. Second, our PMCM can mechanistically explain how discrete categories of risk attitudes (i.e., risk-averse, risk-neutral, risk-seeking) can be understood as a continuum emerging from the interplay between subjective beliefs about potential payoffs and noisiness of perception and can predict corresponding changes in risk attitude across different contexts. Third, we present neuroimaging evidence that fully endogenous fluctuations in the noisiness of *neural* representations of economic variables[55] determine risk attitude in line with a Bayesian perceptual account. This last finding also highlights the promise of using advanced neuroimaging data and analysis methods to help validate and refine computational models of economic choice[104]. Future work in this direction may, for example, link economic preferences as inferred from choice behaviour to the noisiness of neural representations of other relevant economic choice variables, like probability[72,96,101], time[105,106], and social constellations[107,108]. This may further our mechanistic understanding of how these variables are systematically incorporated into choice and should thus inform further development of the corresponding economic models[4,42,107,109–111].

Finally, our results also have potential clinical and policy implications. First, the abundant preference for gambling has long been a puzzle for economists[112,113]. Might the preference for gambling be explained as a misperception of the potential payoffs? One relevant finding in pathological gambling is increased arousal during gambling situations[114,115], which is known to interact with perceptual biases[116]. Indeed, gamblers report unrealistically optimistic beliefs about potential outcomes[117–119]. Future work might investigate false beliefs in pathological gamblers using explicit Bayesian models of risk perception[120,121]. Such false beliefs might have important policy and moral implications for the public communication about gambling. Our results also have implications for the domain of finance. In personal banking, it has become common practice to elicit risk preferences when taking on new clients[122]. If perceptual effects partly drive such elicited preferences, these estimations might be erroneous and clients might be offered an investment portfolio that is overly defensive, just because they have relatively noisy numerical perception or overly pessimistic beliefs about potential long-term stock returns.

To conclude, our work reveals that perceptual biases to presentation order and relative magnitudes, as well as brain state, can shape an individual's risk attitude, and we introduce an experimental and modelling approach that can be used to systematically investigate such effects for a whole range of economic choices. Our findings empirically substantiate emerging proposals that economic choice behaviour should be understood as the fluctuating output of a capacity-limited brain rather than fixed as-is economic preferences[16,25,27,102,103]. Our new model and experimental approach is highly relevant to (neuro)economic research because it formalises why there might be no such thing as an 'objective' risk attitude that is completely independent of context and brain state[6,123], and because it allows researchers to quantify the corresponding effects. More generally, our study demonstrates that attempts to elicit (risk) preferences need to consider the substantial variability in responses and should be mindful of potential confounds like presentation (or fixation) order and the general range of options offered during an experiment. These considerations can substantially impact real-world practice as well, for example when risk elicitation paradigms are increasingly used to compile investment portfolios[122] or characterise the symptoms of psychiatric disorders[124,125].

Methods

Participants and ethics

Thirty right-handed participants (14 females, ages 20–34) volunteered to participate in this study. We informed them about the study's objectives, the equipment used in the experiment, the data recorded and obtained from them, the tasks involved, and their expected payoffs. We also screened participants for MR compatibility prior to their participation in the study. No participant had indications of psychiatric or neurological disorders or needed visual

correction. The study was not designed to investigate sex or gender differences, and no sex- or gender-based analyses were performed. All participants gave written informed consent before taking part, in accordance with the Declaration of Helsinki, and were free to withdraw at any time. The study protocol was approved by the Cantonal Ethics Committee of Zurich.

Procedure

Participants were invited for a total of 4 sessions. Two sessions took place at the 3T scanner and two sessions took place at the 7T scanner (see 'MRI parameters' for more information on scanner type and scanning parameters). Participants always first performed two sessions at one scanner, before the other two at the other scanner. Seventeen participants performed the first two sessions at the 7T scanner, the other 13 at the 3T scanner. During the first and third session—so both at the 3T and 7T scanner—participants performed a calibration version of the risky choice task so we could estimate their average risk attitude and choice consistency within a specific scanner environment before they performed the main task of interest. The calibration task was done in the scanner while undergoing anatomical scans. Participants made choices about 96 possible risky gambles (the safe payoff was either 5, 7, 10, 14, 20, or 28; the risky option was $2^{h/4}$ times the safe option with h being all integers between 1 and 8). These offers were presented twice, once with the safe option first and once with the risky option first. A psychometric probit model with "chose risky option" as the dependent variable and an intercept and the log-ratio of the risky and safe option[27] as independent variables was fitted to these calibration data. The fitted model established the precise mapping between risky/safe payoff ratios and the proportion of risky choices. We then made a participant-tailored design with 8 fractions equally spaced in log-space that—according to the psychometric model—were predicted (equally-spaced) proportions of risky choices between 20% and 80%. These 8 fractions were combined with 6 safe payoffs (5/7/10/14/20/28). Also, all these offers were presented twice with the safe option first and twice with the risky option presented first (so, for every session, each possible safe/risky/order-permutation occurred twice). This amounted to 192 trials per session. After the anatomical scans and the calibration task, participants performed a numerosity mapper task following Harvey et al.[43]. These data were not analysed in the present study.

For the second session in a scanner, participants filled in all information and consent forms and then immediately performed 192 trials of the risk task in the scanner, spread out over 8 blocks of approximately 5 min each.

After all four sessions, we selected a random risky choice trial from each of the two calibration sessions and one of each of the main task sessions. For the risky trials, a digital random number generator was used to hand out the risky option 55% of the time. The participants got the average payout across the 4 sessions on top of their hourly fee (30 CHF per hour; approximately 30–35 USD).

Experimental paradigm

The task of the participants was, for every trial, to choose between a certain amount of money (5/7/10/14/20 or 28 CHF) or a gamble with 55% probability of winning a larger amount of money and a 45% probability of winning nothing at all. The choices were presented by a sequence of tailored stimuli. The general sequence is illustrated in Fig. 1a. The screen always contained a red cross with two diagonal lines to keep fixation near the centre of the screen and to not confound the numerosity of stimuli like a standard fixation cross or point might[43]. The start of a new trial was indicated by the fixation cross turning green for 250 ms. Then, after a pause of 300 ms with just a red fixation cross, a pie chart with a diameter of 1 degree-of-visual-angle (dova) was presented for 300 ms to indicate the probability-of-payout for the coming stimulus. This was always either 55% or 100%. Then, after another 500 ms of just fixation cross, a stimulus array of 1-CHF coins appeared that represented the potential payoff of the first choice option. The coins had a radius of 0.3 dova (degrees of visual angle) and were all randomly positioned with their centre within a circular aperture with a diameter of 5.25 dova. The coin stimulus array was presented for 600 ms, after which only the fixation cross was presented for a jittered duration of either 5, 6, 7, or 8 s. Then, a pie chart indicating the probability of the second payoff was presented for 300 ms, followed by a 300 ms fixation screen, and another coin stimulus array, representing the potential payoff of the second option, again for 600 ms. As soon as the second stimulus array was presented, participants could indicate their response with their index (first-presented option) or middle finger (second-presented option). As soon as they responded, they saw a 1 or 2 indicating which option they had chosen for 500 ms. After the coin stimulus pile was presented, the remaining duration of the trial was either 4, 4.5, 5, or 5.5 s. During the time that was left after the feedback stimulus, participants had to indicate "how certain" they were about their choice using a 4-step Likert scale (by pressing response buttons with four fingers, from index finger-pinky). The results from this certainty rating are not presented in this paper.

At the 7T scanner, stimuli were presented using the Avotec (Avotec, FL, USA) binocular goggle system (800 × 600) with a field-of-view of approximately 25 dova.

Behavioural analyses

Psychometric/KLW model

Model specification. Khaw et al.[27] showed that in risky choice paradigms where participants choose between a certain payoff C and a risky payoff X with a probability of payout p, observed choice behaviour can be understood as the outcome of a Bayesian inference process.

Specifically, their model (from here on referred to as KLW model) assumes that participants do not make decisions based on the objective payoffs C and X, but based on noisy representations of those payoffs, described as random variables r_x and r_c. The likelihood function describes the probability of noisy representations conditional on C and X:

$$r_x \sim N(\log X, \nu^2), \quad r_c \sim N(\log C, \nu^2) \tag{1}$$

As can be seen in the formula, these representations take place in a logarithmic payoff space. This means that participants will adhere to Weber's law. Thus, in natural space, the randomness of their choice behaviour is predicted by ratios between options rather than absolute differences in natural space. Moreover, the parameter ν determines the noisiness of representations and thereby how variable behaviour will ultimately be.

Crucially, the KLW model also assumes that people apply a *common prior* for both the risky and safe options:

$$\log X, \log C \sim N(\mu, \sigma^2) \tag{2}$$

In the KLW model, the μ-parameter has no influence on predicted behaviour. However, because the decision-maker applies the same common prior to both risky and safe options, risky options will always *be relatively underestimated compared to the safe option* (this is because risky options always have higher payoffs than safe options for them to be attractive to the participants). The extent to which underestimation of larger payoffs happens is a function of the ratio between the variance of the likelihood ν^2 and the dispersion of the prior σ^2, β:

$$\beta = \frac{\sigma^2}{\sigma^2 + \nu^2} \tag{3}$$

Specifically, the conditional expected values of the two options conditional on r are:

$$E[X|r] = e^{\alpha + \beta r_x},\ E[C|r] = e^{\alpha + \beta r_c} \quad (4)$$

where α is a function of the mean and standard deviation of the prior:

$$\alpha = (1 - \beta)[\mu + (1/2)\sigma^2] \quad (5)$$

(Rational) participants will choose the risky option if and only if $p \times E[X|r] > E[C|r]$ or, equivalently, $logp + \beta r_x > \beta r_c$, or $logp + \beta r_x - \beta r_c > 0$. Since both r_x and r_c are (independent) Gaussian random variables, conditional on X and C, their difference is a Gaussian variable as well:

$$r_x - r_c \sim N(logX - logC,\ 2\nu^2) \quad (6)$$

Crucially, this means that the probability that the participant chooses the risky option ($logp + \beta r_x - \beta r_c > 0$) is conditional on C and X can be described using the cumulative normal distribution $\Phi(x)$:

$$\Phi\left(-\frac{\beta^{-1}logp^{-1} - log(X/C)}{\sqrt{2}\cdot\nu}\right) \quad (7)$$

The likelihood of the KLW model is equivalent to that of a generalised Linear Model (GLM) with the Normal CDF Φ as the link function and a Bernoulli error distribution (probit model). Such a GLM can be fitted using off-the-shelf methods in most standard statistical modelling software. When using a GLM-approach, the choice of the risky option is the dependent variable and an intercept (all 1's) regressor and $log(X/C)$ as a regressor. Crucially, the resulting intercept parameter δ and slope parameter γ from the probit model can then be re-expressed in the parameters of the KLW model:

$$\gamma = \frac{1}{\sqrt{2}\nu}, \delta = \frac{\beta^{-1}logp^{-1}}{\sqrt{2}\nu} \quad (8)$$

Parameter estimation. We fitted the psychometric model using the Bayesian hierarchical GLM-estimation package Bambi[126]. The standard probit model assumes that choices are determined by a fixed effect of both an intercept (determining the risk aversion) and the log ratio of the risky and safe option. We also estimate participant-wise random effects on all parameters so we had participant-wise KLW parameters:

$$p(risky_choice) \sim \Phi\left(\beta_0 + \beta_1 log\frac{X}{C} + \left(\beta_{2,i} + \beta_{3,i} log\frac{X}{C}\right)\right) \quad (9)$$

Where β_0 and β_1 are fixed effects and $\beta_{2,i}$ and $\beta_{3,i}$ are random (participant-specific) effects. Or, in the formula-syntax of lme4/bambi:

$$risky_choice \sim 1 + log_risky_safe + (1 + log_risky_safe \mid participant_id)$$

For some analyses in which we wanted to see how risk attitudes are modulated for different contexts, we also included the order and/or the value of the safe option in the model as independent variables, so:

$$risky_choice \sim 1 + log_risky_safe * order * n_safe + (1 + log_risky_safe * order * n_safe \mid participant_id)$$

Note that the point-of-indifference of the probit model is a function of both the intercept δ and slope parameter γ ($-\frac{\delta}{\gamma}$). To meaningfully quantify the point-of-indifference, we used the risk-neutral-probability (RNP)[27,35]. This is the hypothetical probability of the risky option for which a risk-neutral decision-maker would show the same point-of-indifference as the participants. Thus, any RNP above 55% implies risk-seeking, whereas any RNP below 55% implies risk aversion. We defined participants as being risk-averse when the 95% Credible Interval of their RNP was below 0.55, and risk-seeking if the CI was above 0.55. When the CI overlapped with 0.55, we categorised participants as risk-neutral.

The risk-neutral probability can be derived from the slope γ and intercept δ as follows:

$$\mathrm{RNP} = \exp\left(-\frac{\delta}{\gamma}\right) \quad (10)$$

To fit the hierarchical probit/KLW model, we used the standard weakly informative priors as implemented by default in the Bambi (based on the approach of the well-known *rstanarm*-package) package[126]. Briefly, the priors are Gaussian distributions centred at 0, with as a standard deviation 2.5 times the ratio between the standard deviation of the dependent and independent variable[126]. To sample from the parameter posteriors, we used the No-U-Turn sampler (NUTS), which is a self-tuning version of the Hamiltonian MCMC sampler and is particularly efficient for high-dimensional models with correlations between the parameter, such as our model here[127]. We collected 4 chains of 2000 samples each (1000 burnin). We always visually inspected the traces of the NUTS sampler and made sure the Gelman-Rubin statistic was below 1.05 for all parameters.

PMCM (Perception and Memory-based Choice Model)

Model specification. Our Perception and Memory-based Choice Model (PMCM) assumes that participants base their risky choices on noisy representations of the payoffs and choose the one with the largest expected payoff on a particular trial. Crucially, in the PMCM, the noisiness of the representation may depend on the order of presentation:

$$r_{x,1} \sim N(log\,X,\ {\nu_1}^2), r_{x,2} \sim N(log\,X,\ {\nu_2}^2) \quad (11)$$

$$r_{c,1} \sim N(log\,C,\ {\nu_1}^2), r_{c,2} \sim N(log\,C,\ {\nu_2}^2) \quad (12)$$

Where $r_{x,1}$ pertains to the representation of the payoff of the risky option presented first, $r_{x,2}$ to the representation of the payoff of a risky option presented second, $r_{c,1}$ to the representation of the payoff of a safe option presented first, and $r_{c,2}$ to the payoff of a safe option presented second. Note that the noisiness of the risky and safe options ν_i is the same, only order of presentation influences noisiness. We expected that the noise for the first option ν_1 will be higher than that of the second option ν_2, but this is not enforced anywhere in the model (e.g., they have identical priors in the estimation procedure).

The PMCM also assumes that participants (potentially) employ different priors for the risky and safe payoffs, because these payoffs come from objectively different distributions:

$$log\,X \sim N(\mu_x,\ {\sigma_x}^2), log\,C \sim N(\mu_c,\ {\sigma_c}^2) \quad (13)$$

For a given parameter set $[\nu_1, \nu_2, \mu_x, \mu_c, \sigma_x, \sigma_c]$ and objective payoffs X and C, we can now obtain the distribution of the expectations of the participants for the two options, which is the product of the prior and evidence distributions and follows a normal distribution (in logarithmic space) :

$$E[logX|r] \sim N\left(r_x + \frac{{\nu_i}^2}{{\nu_i}^2 + {\sigma_x}^2}\cdot(\mu_x - r_x), {\nu_i}^2\right) \quad (14)$$

$$E[logC|r] \sim N\left(r_c + \frac{{\nu_i}^2}{{\nu_i}^2 + {\sigma_c}^2}\cdot(\mu_c - r_c), {\nu_i}^2\right) \quad (15)$$

Thus, the distribution of differences between these expectations in log space is given by the difference of these two distributions, $E[logX|r]$ - $E[logC|r]$, which is also a normal distribution:

$$E[logX|r] - E[logC|r] \sim N(\delta, \varsigma^2) \tag{16}$$

with

$$\delta = r_x + \frac{{\nu_i}^2}{{\nu_i}^2 + {\sigma_x}^2} \cdot (\mu_x - r_x) - \left(r_c + \frac{{\nu_i}^2}{{\nu_i}^2 + {\sigma_c}^2} \cdot (\mu_c - r_c)\right) \tag{17}$$

and

$$\varsigma^2 = {\nu_1}^2 + {\nu_2}^2 \tag{18}$$

The likelihood of choosing the risky option according to the PMCM is defined as the probability that the participant's estimate of the difference between the risky and safe option ($E[logX|r] - E[logC|r]$) is larger than the payout probability of the risky option in log space:

$$p(E[logX|r] + logp > E[logC|r]) = \Phi\left(\frac{logp + \delta}{\varsigma}\right) \tag{19}$$

Where $\Phi(x)$ is the standard cumulative normal distribution.

Model estimation. We implemented the PMCM using the Bayesian statistical modelling library pymc (v.5)[128] and wrapped it in our Python package *bauer* (https://github.com/ruffgroup/bauer/). We used a hierarchical (regression) approach for estimation. This means that for all main six parameters of the model [ν_1, ν_2, μ_x, μ_c, σ_x, σ_c], we assume a group distribution which is a Gaussian distribution. Furthermore, for the ν- and σ-parameters, which are necessarily non-negative, we estimated transformed parameters in an unrestricted space [−∞, ∞] which we then transformed into the non-negative domain [0, ∞] using the softplus function $log(1 + \exp(x))$.

For a given parameter θ (e.g., ν_1 or σ_x), the participants-specific parameter of participant p was a linear combination of the mean group parameter μ_θ plus an individually determined offset-parameter δ_θ times the standard deviation of the group distribution σ_θ:

$$\theta_p = \mu_\theta + \delta_\theta \cdot \sigma_\theta \tag{20}$$

This offset-based specification of the hierarchical model was chosen to prevent the likelihood "funnels" that plague high-dimensional hierarchical models[129].

When relating the PMCM parameters to neural measures, we used a regression approach[65], where a given parameter θ_p (e.g., ν_1 or σ_x) of participants p on trial t was a linear sum of both an intercept θ_0 and a regression coefficient θ_n times the neural measure at trial t, n_t:

$$\theta_{p,t} = \theta_{p,0} + \theta_{p,n} \cdot n_{p,t} \tag{21}$$

We used mildly informative priors on all group distribution parameters:

For the evidence parameters:

$$\mu_{\nu_1}, \mu_{\nu_2} \sim N(-1, 0.5) \tag{22}$$

(before softplus transformation)

For the mean of the prior on risky payoffs:

$$\mu_{\mu_x} \sim N(\hat{\mu}_x, 0.5) \tag{23}$$

Where $\hat{\mu}_x$ is the *objective* mean of the payoffs of risky options (in log space)

For the mean of the prior on safe payoffs:

$$\mu_{\mu_c} \sim N(\hat{\mu}_c, 0.5) \tag{24}$$

Where $\hat{\mu}_c$ is the *objective* mean of the payoffs of safe options (in log space)

For the standard deviation of the priors:

$$\mu_{\sigma_x}, \mu_{\sigma_c} \sim N(0, 0.5) \tag{25}$$

(before softplus transformation)

The regression coefficient on (z-scored) neural measures was a Gaussian centred at 0 with a standard deviation of 1:

$$\mu_{\theta_n} \sim N(0, 1) \tag{26}$$

The prior on all group variance parameters was a half-Cauchy distribution[130]:

$$\sigma_{\nu_1}, \sigma_{\nu_2}, \sigma_{\mu_x}, \sigma_{\mu_c}, \sigma_{\sigma_x}, \sigma_{\sigma_c}, \sigma_{\theta_n} \sim \text{half} - \text{Cauchy}(0.25) \tag{27}$$

Posterior estimates were obtained using the NUTS-sampler, as implemented in pymc 5, with a target acceptance rate of 90%. We obtained 4 chains with 3000 samples each (1500 burnin). If we encountered divergences, we increased the target acceptance rate to 92.5%.

To obtain estimates of the size and presence of an effect, we used "Bayesian" p values –the probability mass of the posterior estimate above/below 0[131]. We generally used one-tailed cutoffs of 5% probability mass.

Model comparisons were performed using the estimated log pointwise predictive density (ELPD[50]) which is a state-of-the-art model comparison technique that—unlike other information theoretic measures like BIC and DIC—takes into account the shape of the posterior distribution and the *effective* number of parameters of the model.

MRI parameters

3T. We acquired functional MRI data using the Philips Achieva 3T whole-body MR scanner equipped with a 32-channel MR head coil, located at the Laboratory for Social and Neural Systems Research (SNS-Lab) of the UZH Zurich Center for Neuroeconomics. We collected 8 runs of fMRI data with a T2*-weighted gradient-recalled echo-planar imaging (GR-EPI) sequence (150 volumes + 5 dummies; flip angle 90 degrees; TR = 2286 ms, TE = 30 ms; matrix size 96 × 70, FOV 240 × 175 mm; in-plane resolution of 2.5 mm; 39 slices with thickness of 2.5 mm and a slice gap of 0.5 mm; SENSE acceleration in phase-encoding direction (left-right) with factor 1.5; time-of-acquisition 4:52 min). Additionally, we acquired high-resolution T1-weighted 3D MPRAGE image (FOV: 256 × 256 × 170 mm; resolution 1 mm isotropic; Shot TR = 2800 ms; TI = 1098.6 ms; 256 shots, flip angle 8 degrees; TR = 8.3 ms; TE = 3.9 ms; SENSE acceleration in left-right direction 2; time-of-acquisition 5:35 min).

7T. We also acquired MRI data using a Philips Achieva 7T whole-body MR scanner (located at the Institute for Biomedical Technology of ETH Zurich) equipped with: 7T MRI quadrature transmit/32-channel head receive array coil (Nova Medical). For the functional data, we acquired T2*-weighted GRE-EPI sequences (150 volumes + 5 dummies); flip angle 74 degrees, TR = 2300 ms; TE = 15 ms; matrix size 109 × 128 (LR × PA); FOV 190 × 224 mm; in-plane resolution of 1.75 mm; 76 slices with thickness of 1.75 mm with no slice gap; SENSE-acceleration in phase-encoding direction (left-right) with factor 3; Multiband acceleration with factor 2; time-of-acquisition 5:35 min). At 7T, we acquired a T1-weighted 3D MPRAGE image with the following parameters: FOV = 240 × 240 × 160 mm; voxel resolution = 0.8 × 0.8 × 0.8 mm; inversion time (TI) = 1656.2 ms; 164 shots; flip angle = 7°; repetition time (TR) =

9.8 ms; echo time (TE) = 4.5 ms; SENSE acceleration factor = 3 (left-right) and 1.5 (anterior-posterior); total acquisition time = 4 min 43 s.

fMRI preprocessing

Preprocessing on fMRI data was performed using fMRIPrep 20.2.2[132], which is based on Nipype 1.6.1[133,134].

Anatomical data preprocessing. The T1-weighted (T1w) images collected from the 3T and 7T data were processed together. All of them were corrected for intensity non-uniformity (INU) with N4BiasFieldCorrection[135], distributed with ANTs 2.3.3[136]. The T1w-reference was then skull-stripped with a Nipype implementation of the antsBrainExtraction.sh workflow (from ANTs), using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and grey-matter (GM) was performed on the brain-extracted T1w using fast (FSL 5.0.9 [137]). A T1w-reference map was computed after registration of 3 T1w images (after INU-correction) using mri_robust_template(FreeSurfer 6.0.1[138]). Brain surfaces were reconstructed using recon-all (FreeSurfer 6.0.1[139]), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical grey-matter of Mindboggle[140]. Volume-based spatial normalisation to one standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with antsRegistration (ANTs 2.3.3), using brain-extracted versions of both T1w reference and the T1w template. The following template was selected for spatial normalisation: ICBM 152 Nonlinear Asymmetrical template version 2009c[141].

Functional data preprocessing. For each of the 24 BOLD runs found per participants (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. BOLD runs were slice-time corrected using 3dTshift from AFNI 20160207[142]. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using mcflirt (FSL 5.0.9[143]). A B0-nonuniformity map (or fieldmap) was estimated based on two (or more) echo-planar imaging (EPI) references with opposing phase-encoding directions, with 3dQwarp[142] (AFNI 20160207). Based on the estimated susceptibility distortion, a corrected EPI (echo-planar imaging) reference was calculated for a more accurate co-registration with the anatomical reference. The BOLD reference was then co-registered to the T1w reference using bbregister (FreeSurfer) which implements boundary-based registration[144]. Co-registration was configured with six degrees of freedom. The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying a single, composite transform to correct for head-motion and susceptibility distortions. These resampled BOLD time-series will be referred to as preprocessed BOLD in original space, or just preprocessed BOLD. The BOLD time-series were resampled onto the following surfaces (FreeSurfer reconstruction nomenclature): fsaverage, fsnative. Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD was computed using two formulations following Power 2014 (absolute sum of relative motions[145]) and Jenkinson (relative root mean square displacement between affines[143]). FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by Power et al.[145]). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction(CompCor[146]). Principal components are estimated after high-pass filtering the preprocessed BOLD time-series (using a discrete cosine filter with 128 s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components are then calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF + WM) are generated in anatomical space. The implementation differs from that of Behzadi et al. in that instead of eroding the masks by 2 pixels on BOLD space, a mask of pixels that likely contain a volume fraction of GM is subtracted from the aCompCor masks. This mask is obtained by dilating a GM mask extracted from the FreeSurfer's aseg segmentation, and it ensures components are not extracted from voxels containing a minimal fraction of GM. Finally, these masks are resampled into BOLD space and binarized by thresholding at 0.99 (as in the original implementation). Components are also calculated separately within the WM and CSF masks. For each CompCor decomposition, the k components with the largest singular values are retained, such that the retained components' time series are sufficient to explain 50% of variance across the nuisance mask (CSF, WM, combined, or temporal). The remaining components are dropped from consideration. The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. The confound time series derived from head motion estimates and global signals were expanded with the inclusion of temporal derivatives and quadratic terms for each[147]. Frames that exceeded a threshold of 0.5 mm FD or 1.5 standardised DVARS were annotated as motion outliers. The BOLD time-series were resampled into standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using antsApplyTransforms (ANTs), configured with Lanczos interpolation to minimise the smoothing effects of other kernels[148]. Non-gridded (surface) resamplings were performed using mri_vol2surf (FreeSurfer).

Many internal operations of fMRIPrep use Nilearn 0.6.2[149], mostly within the functional processing workflow. For more details of the pipeline, see the section corresponding to workflows in fMRIPrep's documentation.

fMRI analyses

The main goal of our fMRI analyses was to get a trialwise estimate of the acuity of the representation of payoff magnitudes in the numerical parietal cortex[35]. Following earlier work[35,40,41], we used an encoding/decoding-modelling approach, where we inverted an encoding model that describes how a voxel i responds to specific stimulus magnitude s $f(s) \rightarrow x_i$ in a Bayesian framework, by extending the encoding model to a multivariate likelihood function $p(X|s)$ using a multivariate t-distribution: $[x_1, .., x_n] \sim f_{1..n}(s) + \epsilon$ with $\epsilon \sim t(0, \Sigma, d)$, where Σ is the residual covariance and d is the degrees-of-freedom of the t-distribution. Using an explicit likelihood function allows us to decode from trial-to-trial what was the presented payoff magnitude, as well as the acuity of the neural response. For every trial, we test the consistency of the BOLD activation pattern for all possible numerosities. The acuity of the neural response was operationalised as the dispersion (standard deviation) of the decoded posterior. Loosely speaking, the posterior will be less dispersed when the BOLD responses of many voxels agree with a small set of numerosities. When the amplitude of the response across voxels is smaller and/or they are not consistent with the same numerosities, the posterior will be more dispersed, as the decoder is less certain about which stimulus was presented. We postulate, as other have done before[40,41], that the uncertainty of our decoding of the presented stimulus is related to the acuity of the *neural representation* of that stimulus.

The main fMRI analysis can roughly be split up in the following steps: (1) Fit a single-trial GLM to estimate trialwise measures of the

response amplitude across voxels, (2) Fit a numerical receptive field model[43] to the response, (3) Fit a multivariate noise model to the residuals of the nPRF model in a leave-one-run-out cross-validation scheme[41], (4) Obtain a posterior estimate of the payoffs magnitudes of unseen data using the noise model and an inverted nPRF model.

Single trial estimates. We used the GLMSingle Python package[150] to obtain single-trial BOLD estimates. Briefly, the GLMSingle package uses cross-validation to do model selection over GLMs with (a) a library of different hemodynamic response functions, (b) different L2-regularisation parameters to shrink the single trial estimates, combating the issue of correlated single trial regressors[151]. (c) GLMSingle also obtains GLMDenoise[152] regressors based on the first n PCA components in a set of noise voxels. Noise voxels are defined as having low explained variance in the task-based GLM. The number of PCA components is selected via cross-validation.

As input to GLMSingle, we used both the 24 first and 24 second payoff presentations per run. For the second payoff presentations, we modelled all trials with the same numerosity as being in the same condition, to aid GLMSingle with cross-validation (similar numerosities should have similar responses). After extensive analysis piloting, we chose to not include any additional confound regressors (e.g., motion parameters, RETROICOR parameters or aCompCorr regressors) in addition to the GLMDenoise confound regressors, as additional regressors did not lead to robustly increased (or even decreased) decoding accuracy and GLMDenoise. This is in line with earlier work on GLMDenoise[152] and the related aCompCorr approach[146].

Numerical receptive field modelling. We fitted a numerical receptive field model to all voxels in the brain, for each session separately (so using 192 single trial estimates). The method is described in detail elsewhere[35,43], so we only briefly describe it here. First, we estimate the mean μ, standard deviation σ, amplitude A and baseline B of a log-normal receptive field for every voxel in the brain separately, to predict the BOLD response of these voxels to different numerosities.

$$f_i(x) = B_i + A_i \exp\left(-\frac{\left(\log x - \log\left(\frac{\mu_i}{\sqrt{1+\sigma_i^2/\mu_i^2}}\right)\right)^2}{2\log\left(1+\sigma_i^2/\mu_i^2\right)}\right) \quad (28)$$

The second part of this equation is a parameterisation of the log-normal probability density function where μ and σ are the mean and standard deviation of the distribution in natural space. Although this parameterisation is somewhat exotic, it is highly useful when plotting the estimated preferred numerosities and their dispersion, for example on the cortical surface.

We first fit the model by using a grid-search: we correlate the single trial estimates for the first payoff stimulus presentation with the predictions of a large grid of 60 μ-s between 5 and 80 and 60 σ-s between 5 and 40. We then estimate A and B using linear least-squares on the best-correlating μ and σ-parameters. Finally, we used gradient descent[153] to refine parameters further. The explained variance R^2 for each voxel is then transformed to *fsaverage*-surface space to compare across participants. We compared the R^2-maps to an ROI of the (right) numerical parietal cortex (rNPC) in *fsaverage* that we drew for an earlier study. We observed that the R^2-maps were tightly related to this ROI (see Fig. 5a) and therefore chose to use this rNPC-mask in all follow-up analyses. The rNPC mask was transformed from *fsaverage*-surface space to individual anatomical space using *neuropythy* (https://github.com/noahbenson/neuropythy).

Voxel selection. One key methodological choice in the encoding/decoding-framework is which voxels to include in the decoding step. In earlier work we chose an arbitrary number of voxels/surface vertices[35]. Although the number of voxels seemed to make little difference in individual decoding accuracy, it remains an arbitrary choice. Here, we chose to use a nested cross-validation scheme to select the number of voxels for each participants/run in a principled way. The method takes into account both that noisy voxels should have an out-of-sample R^2 lower than 0 and different brains have different sizes. Specifically, when decoding run i (out of a total of 8 runs per session), we would fit the nPRF model on all voxels within the NPC mask on (6-run) subsets of the remaining 7 runs, by leaving another run j out $R_{_i,j}$. Then, for each of these nested folds $R_{_i,j}$, we calculated the *out-of-sample* R^2 on run j for each voxel. Thus, for each voxel and each decoded test run i, we had 6 out-of-sample R^2s, which we averaged over. For decoding run i, we would only use voxels that had a mean out-of-sample R^2 larger than 0.

Decoding. After voxel selection, we used a leave-one-run-out cross validation scheme where the nPRF model was fitted to all runs but the test run, after which also a multivariate noise model was fitted to the residual signals. Specifically, we fitted the following covariance matrix:

$$\Sigma = \rho\tau\tau^T + (1-\rho)I \circ \tau\tau^T + \sigma WW^T \quad (29)$$

as well as the degrees-of-freedom of a 0-centred, multivariate t-distribution. Briefly, τ is a vector as its length the number of voxels (n) in the ROI. It pertains to the standard deviation of the residuals of each voxel. Thus, ρ determines to which extent all voxels correlate with each other ($\tau\tau^T$ is the covariance matrix of perfectly correlated voxels, whereas $I \circ \tau\tau^T$ corresponds to a perfectly diagonal matrix/spherical covariance). W is a square matrix of $n \times n$. Each element $W_{i,j}$ is the product of the receptive fields of voxels i and j across stimulus space $(f_i(S)f_j(S)^T$ with S being the entire stimulus space [5, 6, .., 112]. Thus, the free scalar parameter σ determines to which extent voxels with overlapping receptive fields have more correlated noise.

Once the noise model is fitted, we can now determine a likelihood function for any multivariate BOLD pattern X and for any numerical stimulus s:

$$p(X|s) = t(X - [f_1, f_2, \ldots, f_n(s)], \Sigma, d) \quad (30)$$

Since we assume a flat prior on all integers between 5 and 112, we can now just evaluate this likelihood on all these integers and normalise the resulting probability mass function (pdf) to integrate to 1. We take the expected value of this pdf to be our estimate of the presented stimulus. We take the standard deviation of the pdf as a measure of the acuity of neural coding. This standard deviation is used to split up trials within participants and payoff-numerosities for the probit model and as input to the PMCM as a linear regressor on all the model parameters.

Statistics and reproducibility

No formal statistical method was used to predetermine sample size; sample sizes are comparable to or larger than those in related within-subject psychophysical and neuroimaging studies and we use two sessions. No data were excluded from the analyses. The order of presentation and the safe/risky payoffs were randomised across trials within each session; participants were not assigned to experimental groups, so no group-level randomisation was performed. The investigators were not blinded, as all manipulations were within-subject and defined per trial by the presentation software.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The raw fMRI data generated in this study have been deposited on OpenNeuro under accession code ds007508 and are publicly available at https://openneuro.org/datasets/ds007508[154]. The behavioural data from the symbolic (Arabic-numeral) risk task are publicly available on figshare (https://doi.org/10.6084/m9.figshare.31400430)[155]. Source data are provided with this paper.

Code availability

All the code used in this paper can be found online on github: https://github.com/ruffgroup/risk_order_experiment (https://doi.org/10.5281/zenodo.20733498)[156]. Our computational cognitive models are implemented in the open-source Python library *bauer* https://github.com/ruffgroup/bauer/tree/main (https://doi.org/10.5281/zenodo.20008827)[157]. The used nPRF model and decoding software algorithms are implemented in the open-source Python library *braincoder* https://braincoder-devs.github.io/ (https://doi.org/10.5281/zenodo.10778412)[158].

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Author contributions

G.d.H. and C.C.R. conceived and designed the study and developed the experimental paradigm. G.d.H. programmed the experimental paradigm, collected the data, and performed all behavioural and neuroimaging analyses and computational modelling. F.H. helped set up the 7T MRI acquisition protocols. M.G. contributed to the study design and data acquisition and to the interpretation of the findings. C.C.R. supervised the project. G.d.H. and C.C.R. acquired funding. G.d.H. wrote the original draft and C.C.R. reviewed and edited the manuscript. All authors read, commented on, and approved the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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